

Automated Review Rating System

1. Project Overview

This project aims to automate the process of generating ratings from textual reviews. It involves developing an AI-based system, specifically utilizing deep learning, that automatically predicts the numerical rating (e.g., 1 to 5 stars) of a product, service, or business based solely on user-generated textual reviews.

Problem Statement: Manual review rating processes are prone to inconsistencies and manipulation, making it challenging for platforms to accurately gauge sentiment at scale.

Primary Goal: To accurately predict star ratings (1-5) from textual reviews.

Secondary Goals: To reduce manual effort, enhance feedback consistency, and provide scalable sentiment analysis for platforms.

Expected Outcomes: A robust system that provides reliable, automated rating predictions, fostering more trustworthy feedback ecosystems and aiding platforms in efficiently understanding customer sentiment.

2. Environment Setup

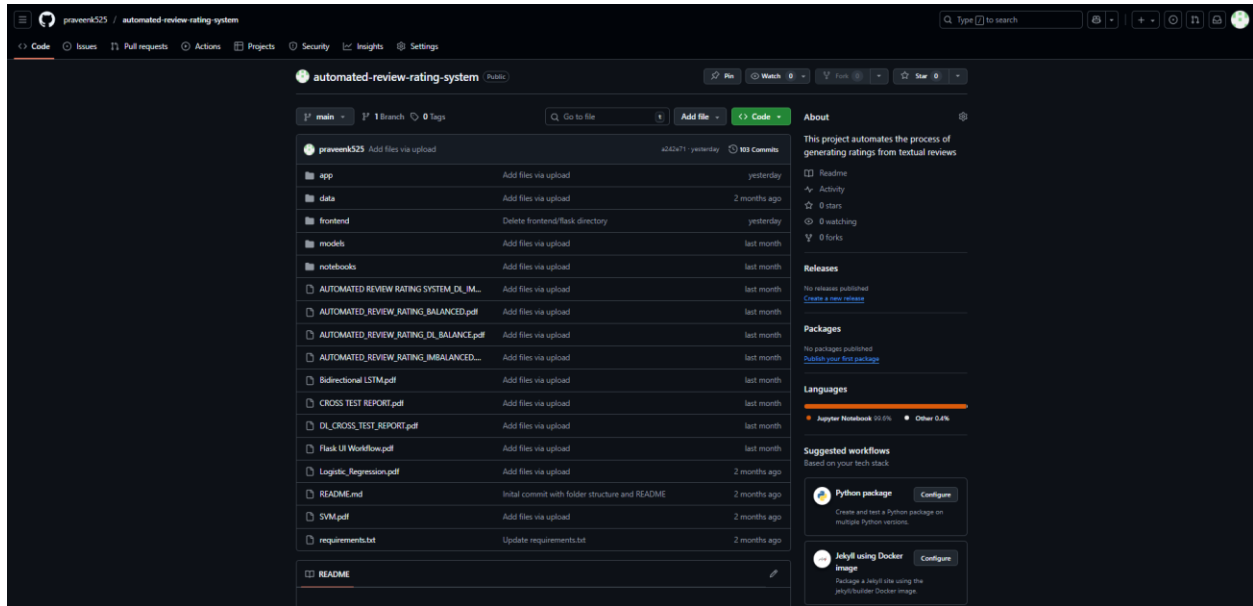
The development environment for this system was primarily Colab, providing access to necessary computational resources. The project was built using Python, with the following key libraries installed and utilised:

- **numpy:** Essential for numerical operations, especially with large datasets and array manipulations.
- **pandas:** Used extensively for data manipulation and analysis, particularly for handling DataFrames.
- **matplotlib** and **seaborn:** Employed for creating various data visualisations to understand data patterns and distributions.
- **scikit-learn:** A comprehensive library providing tools for machine learning tasks, including data splitting and potentially some preprocessing utilities.
- **nltk (Natural Language Toolkit)/Spacy:** Crucial for natural language processing tasks, such as text cleaning, tokenisation, and potentially stopword removal or lemmatisation.

(Note: Specific version numbers for Python and each library are recommended for full reproducibility but were not explicitly detailed in the provided sources.)

3. GitHub Project Repository

The project is hosted on a GitHub Project Repository. *(Note: A direct link to the repository, its high-level directory structure, and any contribution guidelines would enhance this section.)*



Github link: [praveenk525/automated-review-rating-system](https://github.com/praveenk525/automated-review-rating-system): This project automates the process of generating ratings from textual reviews

4. Data Collection

Data for the Automated Review Rating System was collected from Kaggle datasets. A total of 4 datasets were gathered, primarily consisting of reviews from Amazon in the Electronics domains. Each dataset contains two main columns: "review " and "rating" (1–5).

The selection of Amazon reviews from these domains provides a rich and diverse set of customer sentiments relevant to common product and media consumption.

Specific datasets collected include:

- **Amazon Electronics Reviews:** This dataset contains approximately 6,906,564 reviews, covering products like phones, laptops, and cameras, capturing a wide range of customer sentiments.
- **Electronics_part0 & part_1:** This dataset contains approximately 3,778,161 reviews & 2,722,700 reviews focusing on electronics products such as phones, laptops, and cameras. It includes key fields like asin, reviewText, overall, category, and summary, capturing a diverse range of customer sentiments.

- **Electronics Reviews:** This dataset contains approximately 3,000,000 reviews, covering electronics products such as phones, laptops, and cameras. It includes fields like overall (rating), vote, verified (purchase status), reviewTime, reviewText, and summary, capturing a wide spectrum of customer opinions.

All 4 collected datasets were subsequently merged into a single dataset for comprehensive processing. This merging process involved concatenating the review and rating columns from each dataset to form a unified corpus.

5. Data Cleaning and Preprocessing

Preprocessing is a crucial initial step to transform raw data into a clean, structured format suitable for machine learning models. The following steps were performed:

- **Load and Inspect Datasets:**
 - All 4 datasets were loaded, inspecting their structure, column names, data types, and sample records.
 - Verification of "review" and "rating" columns was performed.
 - Functions like `pd.read_csv()` and `pd.read_excel()` were used for loading.
 - Commands such as `df1.head()`, `df1.tail()`, `df1.columns`, `df1.shape`, `df1.dtypes`, `df1.info()`, and `df1.describe()` provided initial insights into the data.
- **Remove Unnecessary Columns:**
 - Columns not required for prediction (e.g. 'overall', 'vote', 'verified', 'reviewTime' etc.) were dropped to simplify the dataset and focus on relevant information.
 - Only review and rating columns were retained.
- **Remove Null or Missing Values:**
 - Rows containing null values in either the "review" or "rating" columns were removed. This step is crucial to prevent model errors and ensure data integrity, as missing values can lead to unexpected model behaviour.
 - `df.isnull().sum()` was used to identify missing values, and `df.dropna(subset=['review_text', 'rating'], inplace=True)` was applied to remove them.
- **Remove Duplicates and Conflicting Reviews:**
 - **Exact duplicates** (same review text and same rating) were handled by keeping only one instance to prevent biasing the model.

- **Conflicting reviews** (same review text but different ratings) were entirely removed. This decision implies that such contradictory entries are considered unreliable or ambiguous, and their inclusion could introduce noise or confusion for the model.
- `df.duplicated().sum()` checked for duplicates, and `df.drop_duplicates(inplace=True)` removed them.
- **Clean Text Data:**
 - All text data was converted to lowercase to standardise text and reduce vocabulary size.
 - The following elements were removed from the text: URLs, punctuation, numbers and special characters, and extra spaces. This ensures that the model focuses on the semantic content of the reviews rather than formatting or extraneous characters.

```
import re

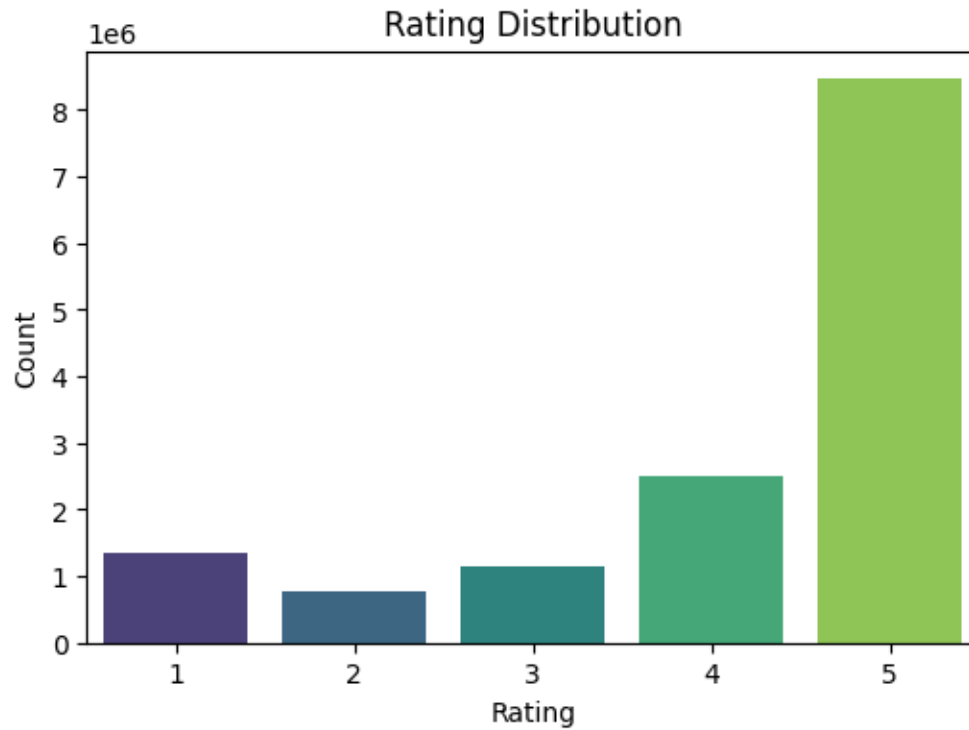
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"^\w\s", "", text)
    text = re.sub(r"\d+", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df.loc[:, "review"] = df["review"].apply(clean_text)
```

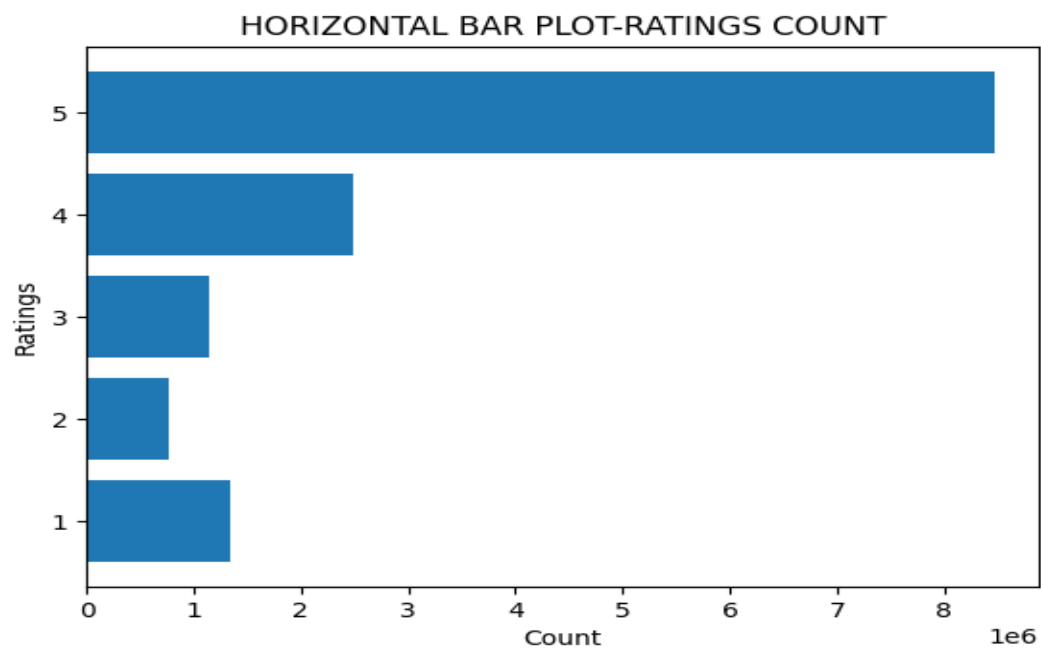
6. Data Visualization

Data visualisation was employed to understand patterns, trends, and insights within the data using visual elements.

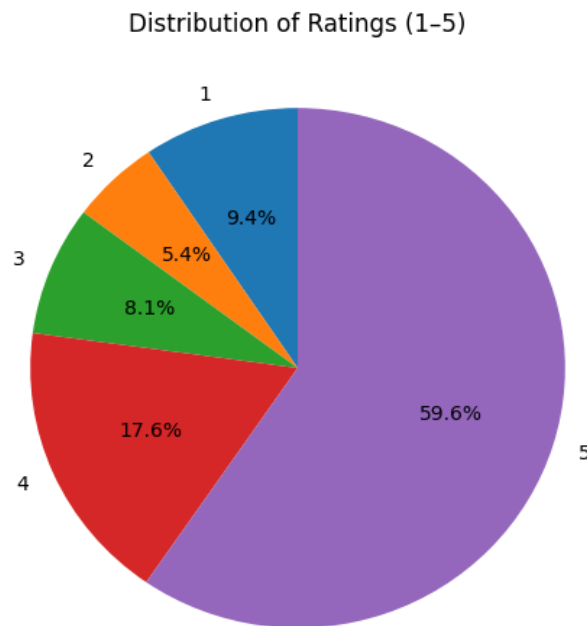
- **Bar Chart:** Used to show the number of reviews for each star rating (e.g., how many 1-star, 2-star, etc.). This helped to identify the distribution of ratings and potential class imbalances.



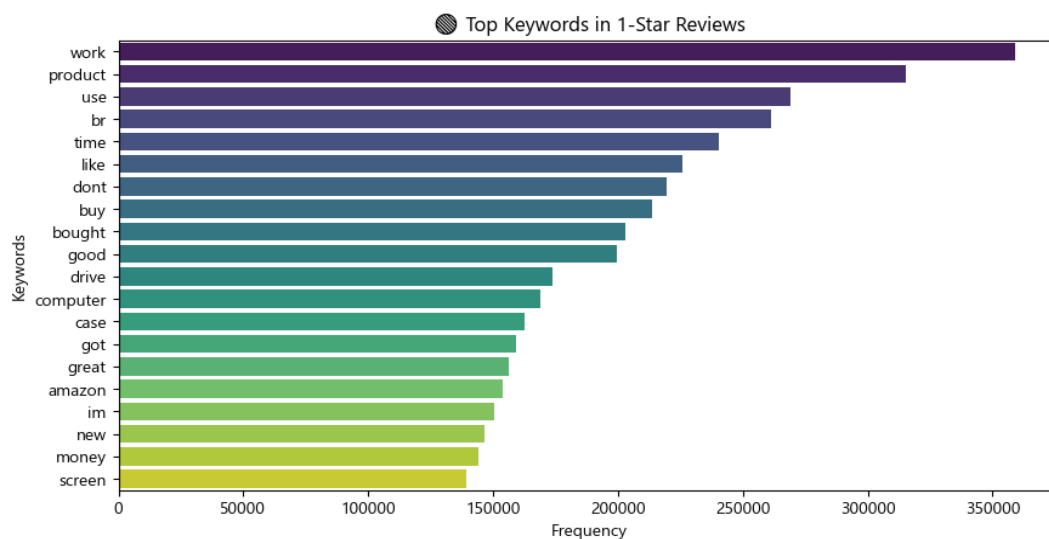
- **Horizontal Bar Chart (HBar):** Used to show average values, such as review length for each rating, in a clean, readable format. This could reveal if, for instance, extremely short or long reviews tend to correlate with specific ratings.

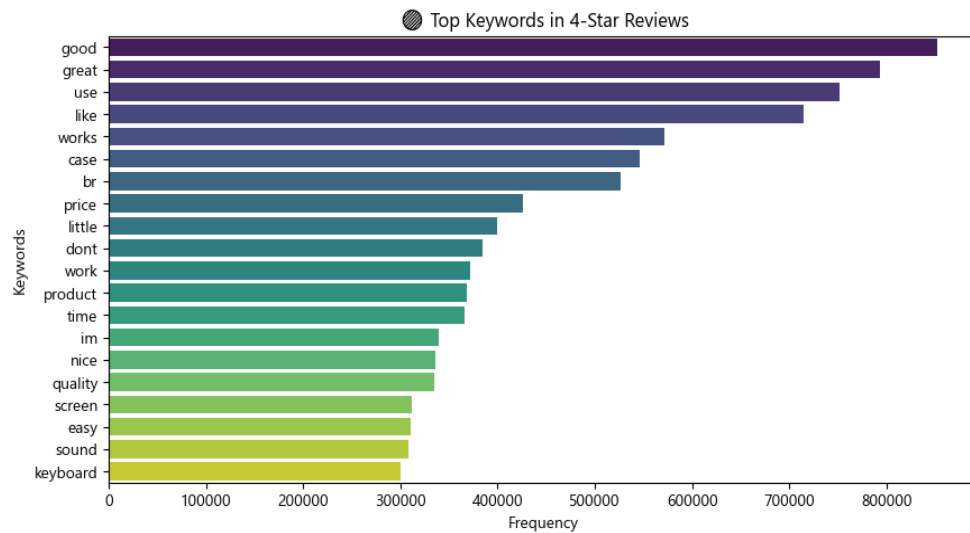
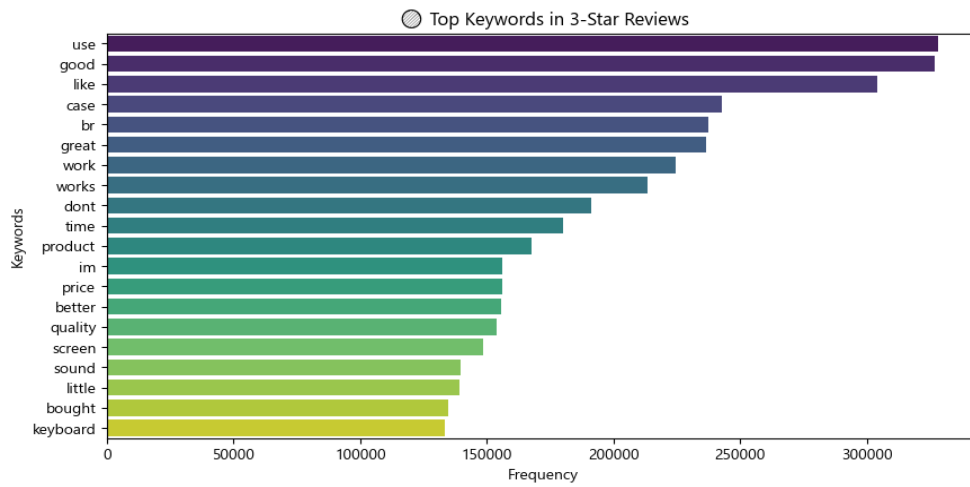
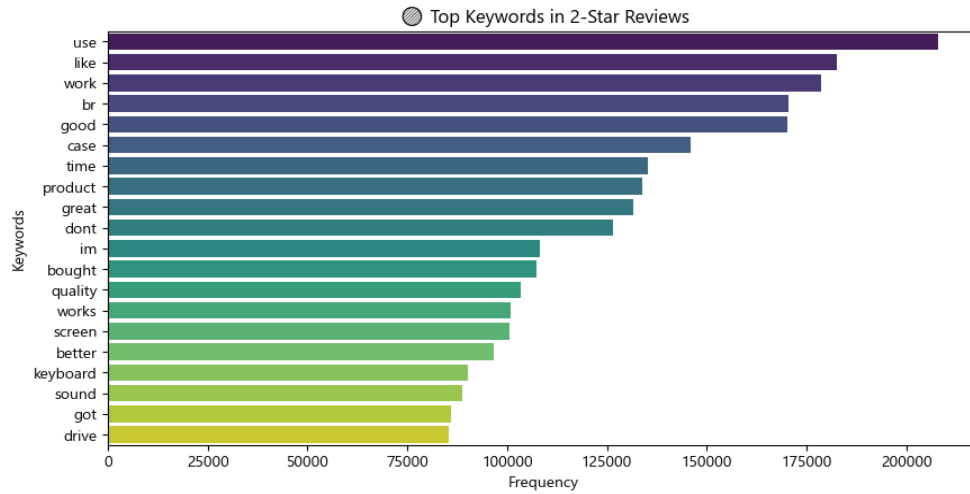


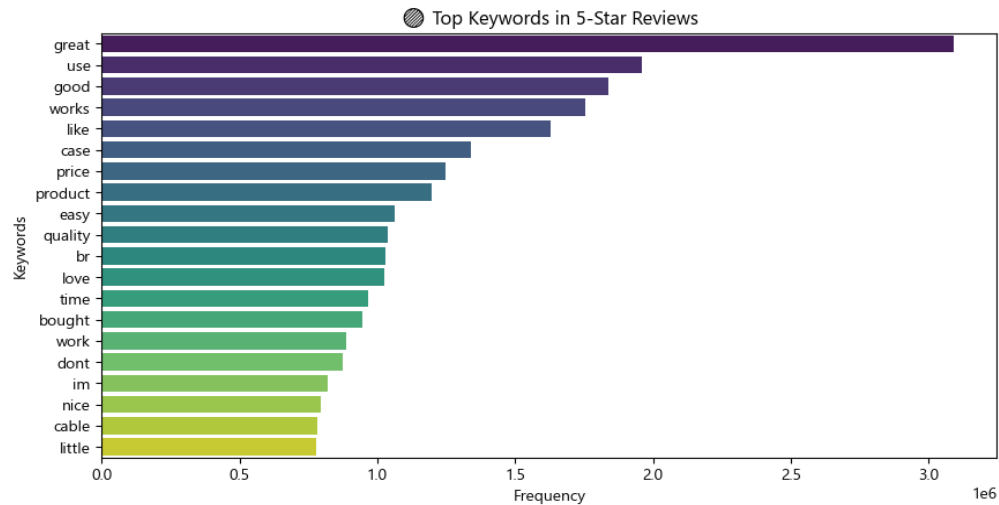
- **Pie Chart:** A circular chart divided into slices to illustrate proportions. Used to show the percentage distribution of each rating (e.g., how much of the data is 5-star, 4-star, etc.). This offered a clear overview of the overall rating proportions in the dataset.



- **Word frequency:** A word frequency visualization in review rating is a graphical representation that shows how often specific words appear in customer reviews, categorized by their rating scores (e.g., 1-star to 5-star). It helps uncover patterns in language usage across different sentiment levels.





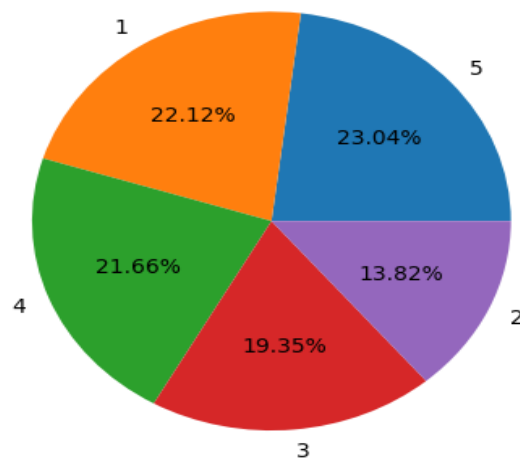


7. Dataset Creation

The initial dataset exhibited imbalanced data, meaning the distribution of classes (star ratings) was not equal, with some classes having significantly more samples than others. This is a common issue in machine learning as models tend to perform poorly on minority classes, leading to biased predictions.

To address this, a balanced dataset was prepared with the following target distribution for ratings:

- **1-star: 480000**
- **2-star: 300000**
- **3-star: 420000**
- **4-star: 470000**
- **5-star: 500000**



8. Samples

The documentation includes sample reviews for each star rating (1-5) to illustrate the nature of the data after preprocessing. These samples provide concrete examples of the textual content and their corresponding ratings, aiding in qualitative understanding of the dataset.

Showing 5 sample review(s) for Rating 1:

1. this charger is not compatible with hp pavillion dm
2. this charger does not charge i liked the design but when i actually tried to charge my iphone and ipod it does nothing other than give the appearance its charging i understand it was cheap and may not have worked as quickly or lasted as long but i would have thought it would have at least worked to some degree
3. screen has a deadstuck pixel will be sending it back for a replacement hope it doesnt disappoint
4. pay for what you get this thing is perfect if youre looking for a quick and dirty solution i went the extra mile and anchored through all the holes i could find on the back even if they werent intended for such things i wish it had more defined slots on the top and bottom instead of just wide rectangular holes
5. i am honestly shocked shocked at the difference between the average rating and what this product deserves the starwritingmills must truly be in full swing so many minor defects that i have to give this a one a no instructions no manual no helpful hints card b not as advertised does not come in a cardboard envelope but rather a syranplastic like envelope c not as advertised does not come with a push card one of those light plastic rectangles like a credit card that help push the cover flat and get rid of bubble instead it comes with a rectangle of thinfilm blue plastic like the stuff that comes on new dishwasher to protect the metal finish you could not use this to apply pressure so i had to use my fingers i thought an actual credit card would be too rough d not as advertised the small square of polishing cloth is about x not useful for anything unless youre the size of a hamster i have a half dozen of these ive gotten for free at an eyeglasssunglass store e i had to use both of the covers included because the first one had dozens of bubbles and im a bit of a pro at installing these f i finally got the nd installed with only bubbles but it actually has a series of fine scratches right in the middle livable but annoying g both covers came with little round plastic shreds stuck to them from the milling process where they cut out the small hole at the top for the camera and the larger hole at the bottom for the function button these leftover milling shreds were as sticky as the cover think about it protection layers the actual cover layers x holes shreds per cover and they kept reappearing and getting stuck under the cover as i was installing it and finally and worst of all these dont cover the whole ipad they leave on each of the long sides unprotected buy at your own peril

Showing 5 sample review(s) for Rating 2:

1. camera filters

2. i bought of these one for my wife and one for me but i didnt realize that the bottom has no covering so if you use it on your lap in bed or someplace other than a desk you can get into the fans my wife used it for about an hour laying on her bed and the blanket would get inside and mess things up so if you are looking for a lap desk this isnt what you want

3. i was very disappointed with this product i thought that i would be able to plug in two i devices although perhaps one would not charge as fast as the other but as it turns out i was only able to charge one i device at a time while this device got great reviews on other websites i was very disappointed when i recently took it on a trip and was only able to charge one i device at a time i have a couple of other dual usb chargers that are capable of charging to two idevices at a time this one was a big disappointment

4. just as the above suggests cheap plastic if your laptop is heavy the thing might bend on you and if you drop it from a mere height of a foot you might break or crack it it takes up your valuable usb port and isnt all the quiet esp if you dropped it or its bent or its on an angle which makes the fan all sound its grinding yeah i ended up swapping the one i had at work which they provided with the one i bought for myself

5. lowepro ridge protects much better

Showing 5 sample review(s) for Rating 3:

1. these dont hold the charge that the oem nikon batteries did i am able to get maybe a few hours instead of days on a fully charged battery better than nothing but shrug

2. unfortunately the tab that tucks under the tablet and secures it inside the case does not have velcro on it another case that i purchased that was similar to this did have velcro and it is extremely handy the tablet tends to slowly work its way out of the case and requires readjustment to get it centered otherwise the construction is sturdy

3. the item arrived demolished a smaller lens on the inner tube had come separated and was bouncing around in the tube against the other lenses for almost a month now the item is so badly damaged and useless i really hope this wasnt intentional i was so excited about using this lens im so disappointed to have to return it just to buy another i wish it would have worked out better but this is the gamble when buying it used on amazon

4. not easy to use it

5. syncs alright with apple and android devices but doesnt show battery capacity on android h of continuous usage and the low battery warning comes on sound quality doesnt have the clarity and bass im accustomed to with beats by monster button controls way more functional and userfriendly than the single buttons that are currently popular

Showing 5 sample review(s) for Rating 4:

1. i bought this monitor about months ago and for the price it works great its nice and bright the color representation seems pretty accurate and after months of heavy use no ghosting or dead pixels theres nothing fancy the design is pretty basic but again for a monitor in this price range i am well pleased i would definitely recommend this to others looking for an inexpensive monitor meant for work related tasks for gaming id suggest something a tad more responsive

2. does perform its function good

3. mine arrived yesterday and i went out this morning for a short stroll i had purchased the gps in anticipation of a coming trip to a very large city where i hope to do some wandering on foot with a high probability of getting turned around as the joke goes im not lost i know where i started and i know where im going and this aint either one of those i installed the batteries after a short tussle with the back of the device and took it outside where i locked onto a signal in something between minutes i did take the advice of other reviews and decided to dispense with the included lanyard which is somewhat insubstantial i plan to get some thin rope at a local marine hardware store and rig up something similar to the one supplied with the backtrack i had a lanyard with a metal connector and decided against that also because of somebodys review about keeping the device away from metal after it started up i put in one marker and took off for a walk of several blocks and continually checked the locator arrow on the dial it always pointed back in the general direction i had started from and when i eventually turned around and headed home over a slightly different route the arrow seemed pretty accurately locked in the correct direction as i got in front of the house the unit went to zero all arrows as its designed to do i was pleasantly surprised to see the backtrack function exactly as advertised and i think i can be quite comfortable using it on the pending trip two things i particularly liked were the fact that the gps gives you a distance measurement which is very helpful keeping a walker advised of his cardio workout and the fact that the compass seems to be pretty accurate in its calibration ive owned a couple of watches with compasses and they required an elaborate calibration procedure on a frequent basis the backtrack seems to have eliminated the need to calibrate and thats a handy extra ralph l seifer long beach california

4. just enough grip

5. works well puts out full power for apple products ran ipad dvd player and ipod touch off it for mile family trip worked great

Showing 5 sample review(s) for Rating 5:

1. great original product at great price fits adonit jot pro

2. i did not realize i was buying a bag full

3. i keep one of these in every room handy and they do the job our phoneipadmac screens

4. i am very pleased with these power line adapters i have a total of in my home connected to pcs connected to an xbox and slingbox via a port hub and one connected to my home server my

download speeds are great and i dont have the hiccups that i used to when i was trying to use wireless throughout the house i notice no lag at all even when multi player gaming on the pc or xbox this a great panasonic product its the next best thing to having your home prewired for networking

5. i tried all the rest this one beats them all even that pricey and thoroughly annoying zagg govt issue helicopter blade protection film blah blah blah shield the zagg film might be great for small surfaces like the iphone but for the ipadfailbr br the ivisor protector starts off with an unbelievably easy installation i read the reviews here and didnt much stock in the claims that it could be that easy boy was i wrong this was a piece of cake install took all of seconds perfect fit no bubbles no sticky edges curling up just a fit like the it was custom made to fit my ipad oh wait it was the engineering and design of ivisor got this right the screen is a just matte enough so there is reduced glare but not enough to affect any picture clarity at all what fingerprints they just dont show up on my screen the tactile quality stays the same my screen is just as responsive as ever to my fingertips or my stylus and so what if the protector gets sticky or oily and needs a good scrub you take it off wash it air dry it and put it back on try to do that with a zagg no dont it wont work it cant be done this protector has stood up under the rigors of grandkids playing games on the ipad one little wipe with a damp cloth and it looks brand new again not a fingerprint scratch or curled edge to be found i am buying more of this great screen protector for my family and recommend them to all my friends with ipads the protection is there the fit is there all the qualities you want ivisor has them winbr br at it is a great investment as part of your protection plan for your or ipad nope i am not a paid poster or a shill just a very happy and satisfied customer who always evaluates the reviews before i busy anything i bought mine from xtreq here quick shipping and excellent service when i needed questions answered

9. Train-Test Split

In machine learning, the dataset was split into:

- Training Set: Used to train the model (e.g., 80% of data).
- Testing Set: Used to evaluate the model's performance on unseen data (e.g., 20% of data). This split helps ensure the model can generalise effectively to new, unseen reviews. The 80-20 split is a commonly used ratio that balances the amount of data available for training with a sufficient portion for independent evaluation.

#stratified train test split

```
X_train,X_test,y_train,y_test=train_test_split(X,y_categorical,test_size=0.2,stratify=y_categorical,random_state=42)
```

10. Bidirectional LSTM (BiLSTM)

The Bidirectional LSTM (BiLSTM) model was selected for this task due to its effectiveness in processing sequential data like textual reviews.

- **Captures context in both directions:** Unlike standard LSTMs, BiLSTM reads the text sequence both forward and backward. This is crucial for understanding sentiment in reviews, where the meaning of words can often depend on their preceding and succeeding context.
- **Effective for sequential data:** Customer reviews are textual sequences. LSTMs are inherently well-suited for handling long-term dependencies within such sequences, significantly outperforming simpler Recurrent Neural Networks (RNNs).
- **Proven performance:** BiLSTM models have demonstrated higher accuracy in sentiment analysis and review rating prediction tasks compared to traditional machine learning algorithms or simpler neural network architectures.

11. Model Training Logic (including BiLSTM-specific Preprocessing)

Before feeding the data into the BiLSTM model, the following additional preprocessing steps were applied, building upon the general data cleaning:

11.1. BiLSTM-Specific Preprocessing Steps:

1. **Text Cleaning:** All text was converted to lowercase, and special characters and punctuation were removed. This ensures a clean input for tokenisation.
2. **Truncating Excessively Long Reviews:** Reviews exceeding 300 words were truncated to reduce noise from overly verbose inputs and ensure a uniform input size for the model.
3. **Encoding Labels:** The star ratings (1–5) were first label-encoded and then converted into one-hot vectors. This is a standard practice for multi-class classification problems when using neural networks with a softmax output layer.
4. **Tokenization and Padding:** Text was tokenized using Keras' Tokenizer, converting words into numerical sequences. These sequences were then padded to a fixed length of `max_len=200`. Padding ensures that all input sequences to the neural network have the same length, which is required for batch processing.
5. **Optional Stopword Removal & Lemmatization:** These steps were considered and applied to further reduce vocabulary size and potentially improve semantic understanding by reducing words to their base form.

11.2. BiLSTM Model Architecture and Training: The BiLSTM model was trained using the following architecture and regimen:

- **Embedding Layer:** Words were converted into dense vector representations. This could be achieved using either randomly initialised embeddings or pretrained embeddings (e.g., GloVe). The choice of pretrained embeddings typically helps the model leverage pre-existing linguistic knowledge.
- **Bidirectional LSTM Layer:** This core layer consisted of 128 units and incorporated dropout and recurrent dropout. The bidirectional nature allows the network to process the sequence from both forward and backward directions, capturing more comprehensive context. Dropout mechanisms help prevent overfitting.
- **Dropout Layer:** A 50% dropout rate was applied after the BiLSTM layer. This technique randomly sets a fraction of input units to 0 at each update during training, which further aids in preventing the model from becoming overly reliant on specific features and improving generalisation.
- **Dense Output Layer:** The final layer comprised 5 neurons with a softmax activation function. This configuration is standard for multi-class classification, where each neuron corresponds to one of the five star ratings, and softmax ensures the output probabilities sum to 1.
- **Loss Function:** categorical_crossentropy was chosen as the loss function, which is suitable for multi-class classification where labels are one-hot encoded.
- **Optimizer:** The Adam optimizer was used for efficient convergence during training. Adam is an adaptive learning rate optimisation algorithm that has proven effective across a wide range of deep learning problems.
- **Callbacks:** EarlyStopping and ReduceLROnPlateau were implemented to dynamically adjust the learning rate and prevent overfitting during training.
 - EarlyStopping monitors a validation metric (e.g., val_loss) and stops training if the metric stops improving for a specified number of epochs (patience).
 - ReduceLROnPlateau reduces the learning rate when a metric has stopped improving, allowing the model to make finer adjustments.

Training was performed using a stratified 80-20 train-test split to ensure that all rating classes were proportionally represented across the training and testing datasets.

12. Evaluation Results and Interpretation

Upon completion of training, the model's performance was evaluated on the unseen test set. The evaluation involved several key metrics:

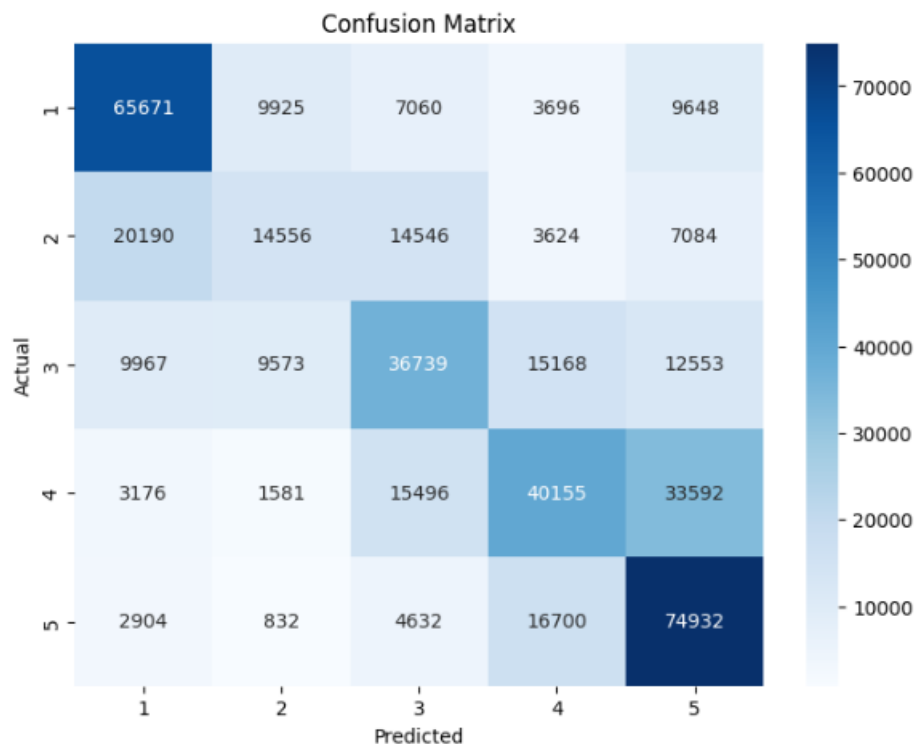
- **Accuracy:** This metric measures the overall proportion of correctly predicted ratings.

- **Precision, Recall, F1-Score:** These metrics were evaluated per class to assess how effectively the model predicted each individual rating (1-5 stars). This is crucial for imbalanced datasets, as high overall accuracy might mask poor performance on minority classes.
- **Confusion Matrix:** A confusion matrix was utilised to visualise misclassifications between different ratings, providing granular insights into specific errors made by the model. It highlights which classes are most often confused with others (e.g., 1-star reviews misclassified as 2-star).

Accuracy: 0.5347

Classification Report:

	precision	recall	f1-score	support
1	0.64	0.68	0.66	96000
2	0.40	0.24	0.30	60000
3	0.47	0.44	0.45	84000
4	0.51	0.43	0.46	94000
5	0.54	0.75	0.63	100000
accuracy			0.53	434000
macro avg	0.51	0.51	0.50	434000
weighted avg	0.52	0.53	0.52	434000



13. Additional Considerations

13.1. Future Work:

- **Explore advanced models:** Investigate Transformer-based models (e.g., BERT, RoBERTa) which often achieve state-of-the-art performance in NLP tasks.
- **Ensemble methods:** Combine multiple models to potentially improve predictive accuracy and robustness.
- **Expand data sources:** Incorporate reviews from a wider variety of domains and languages to enhance generalisation.
- **Hyperparameter optimisation:** Conduct more rigorous hyperparameter tuning to find optimal model configurations.
- **Explainability:** Implement techniques to understand *why* the model makes certain predictions, improving trust and interpretability.

13.2. Conclusion:

This project successfully developed an AI-based system for automated review rating using a Bidirectional LSTM architecture. Through meticulous data collection, cleaning, and specific preprocessing steps, the system aims to provide an efficient and standardised method for sentiment analysis from textual reviews. While the core methodology is robust, further refinement and detailed performance reporting will enhance its practical application and scientific contribution.